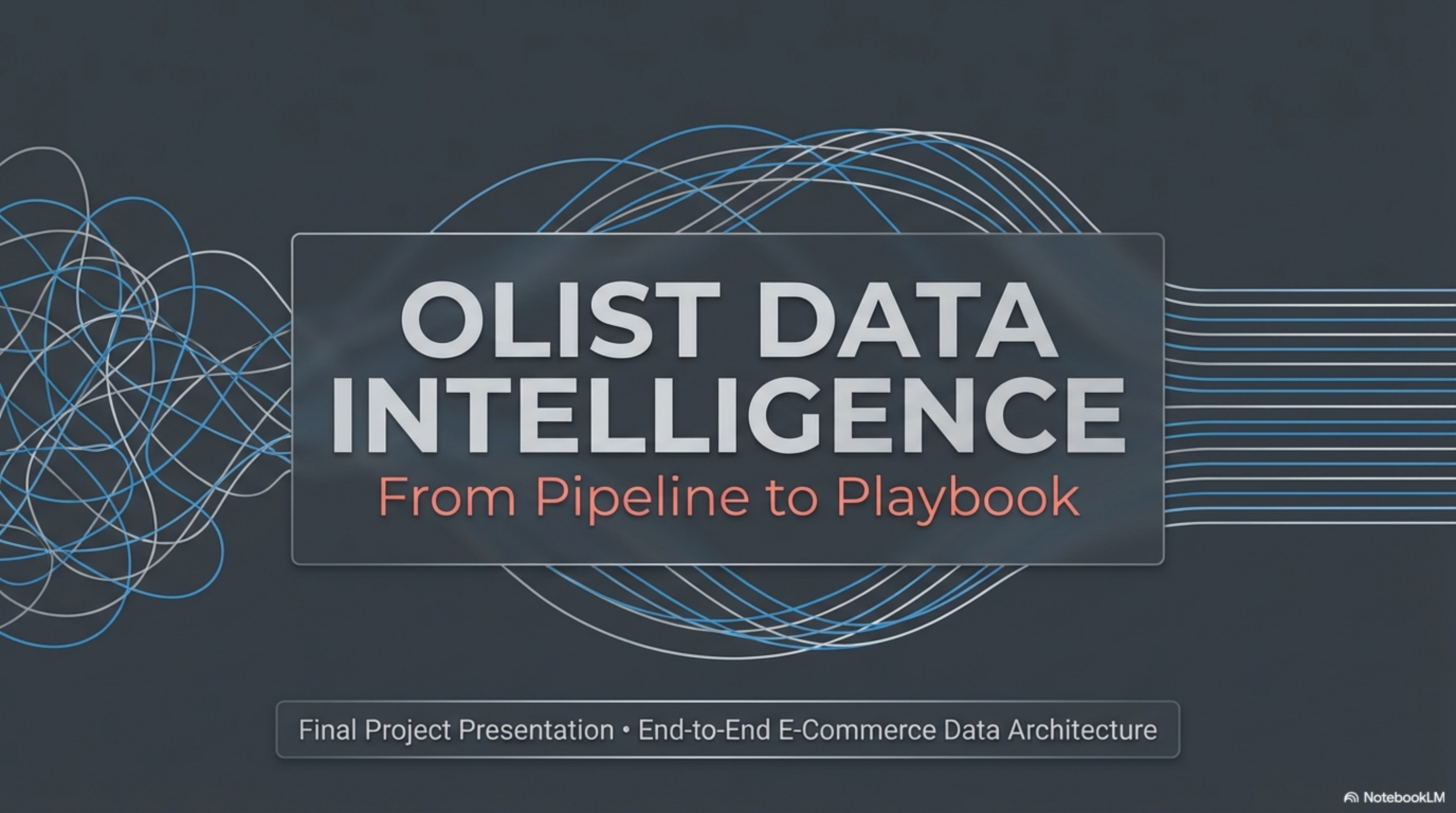




OLIST DATA INTELLIGENCE

From Pipeline to Playbook

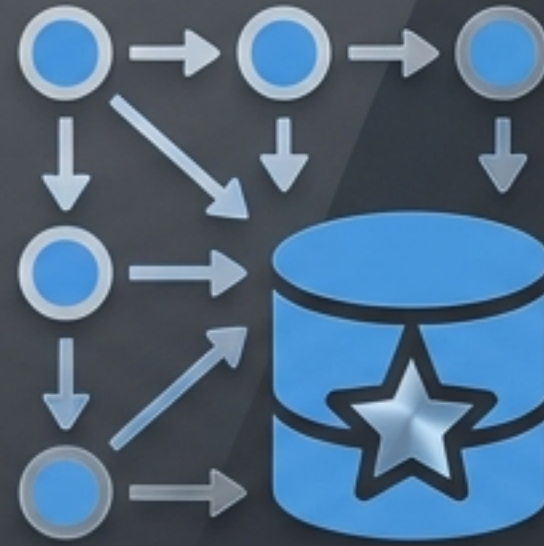
Final Project Presentation • End-to-End E-Commerce Data Architecture

Transforming Raw Commerce Data into Strategic Clarity



Raw & Siloed Data

Over 100,000 unrefined order records, fragmented across isolated CSV files with inconsistent formatting and zero analytical readiness.



Scalable BigQuery Pipeline

An end-to-end, automated ELT pipeline transforming raw inputs via DuckDB into a high-performance Star Schema within Google BigQuery.



Actionable Growth Blueprints

Trustworthy, zero-latency business intelligence that maps market demand, identifies cash-flow drivers, and dictates clear e-commerce venture strategy.

Architectural Decisions Driven by Business ROI

Technical Capabilities

Automated ELT via Python & SQLAlchemy

Eliminates manual data wrangling, allowing analysts to focus strictly on strategic forecasting.



Cloud-Native Data Warehouse (Google BigQuery)

Infinite scalability without infrastructure overhead.



Dimensional Star Schema Design

Multi-dimensional ROI slicing.



Direct Business Value



Zero-latency reporting.

Allows analysts to focus strictly on strategic forecasting.



Infinite scalability without infrastructure overhead.

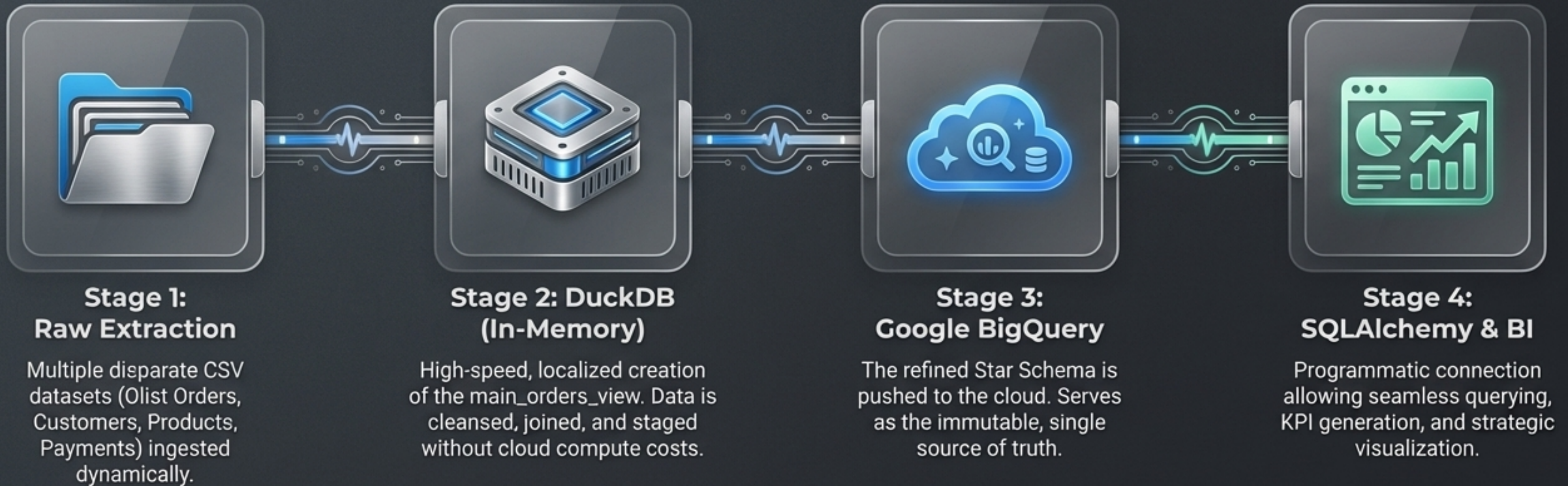
Effortlessly handles seasonal volume spikes with predictable cost structures.



Multi-dimensional ROI slicing.

Enables instant pivoting of data by customer segment, product category, or geographic region for targeted marketing.

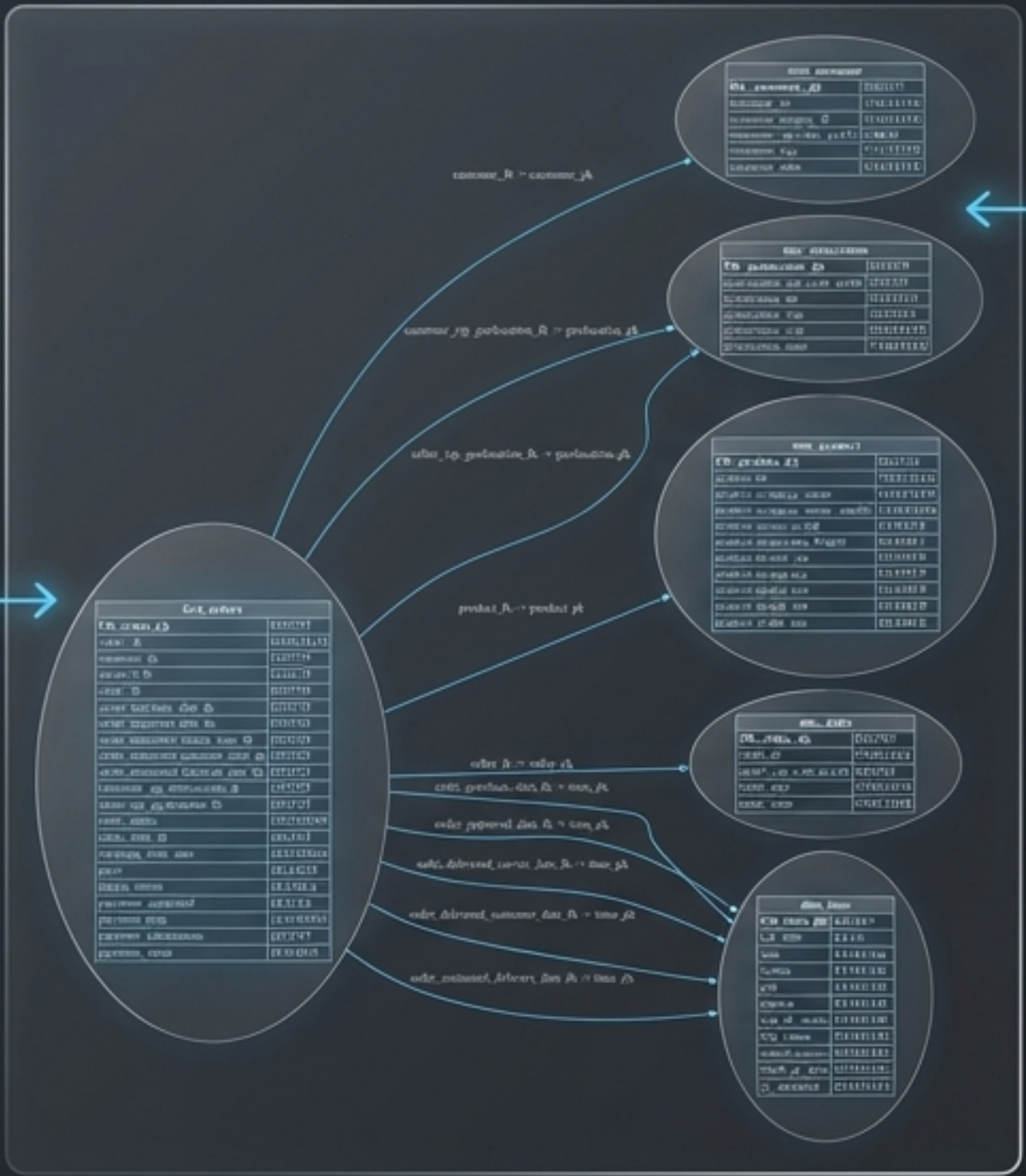
The Data Engine: Ingestion, Transformation, and Analytics



Purpose-Built Star Schema for Accelerated Intelligence

The Central Hub

Records every quantifiable transaction (price, freight, payments) with unique surrogate keys generated via MD5 hashing.



The Analytical Spokes

Denormalized attributes allow business users to filter and group by product category, geographic location, or time without complex, slow database joins.

Why This Matters: This architecture drastically reduces query times from minutes to milliseconds, empowering live, interactive business dashboards.

Uncompromising Data Quality and System Trust

Data Trust Scorecard

Volumetry Reconciliation

Local DuckDB vs. Cloud BigQuery record counts.

[Status: PASS]

Null Validations

Verifying zero nulls in critical order_id fields within fact_orders.

[Status: PASS]

Uniqueness Enforcement

Duplicate checks on primary keys in dimension tables.

[Status: PASS]

Referential Integrity

Mapping fact table foreign keys to dimension primary keys.

[Status: RESOLVED]

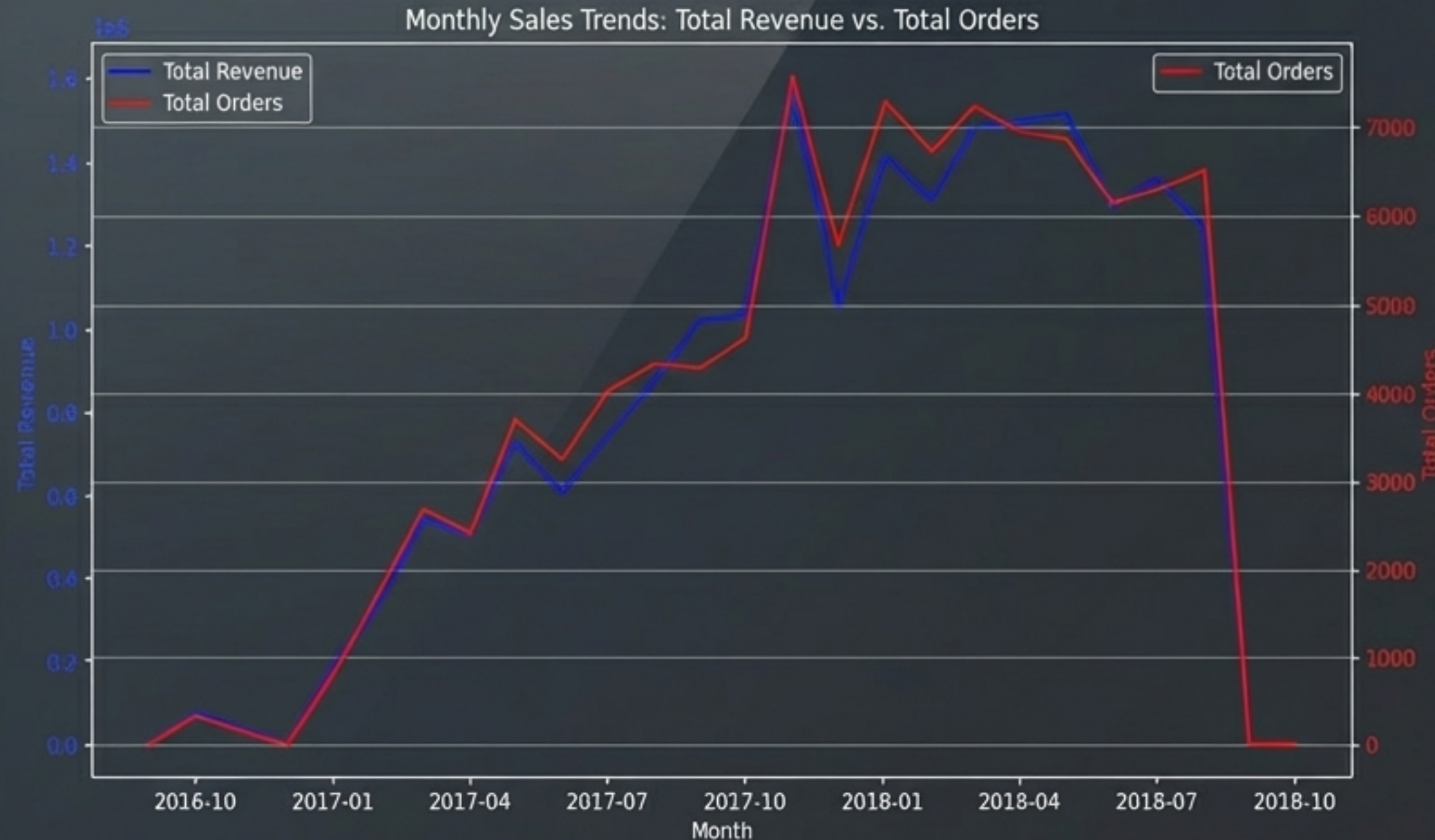
Proactive Orphan Record Remediation

Initial testing revealed 830 fact_orders records mapped to missing products in the source data.

Instead of dropping valuable revenue data, we proactively engineered an 'Unknown' dimension entry (product_pk = 0).

Impact: Preserved 100% of financial transaction data while maintaining strict database integrity.

Macro Growth: A Thriving, Predictable Market



Parallel Scaling

Revenue and Order Volume scale in near-perfect lockstep. This proves that growth is driven by genuine user adoption, not just isolated price inflation.

Predictable Seasonality

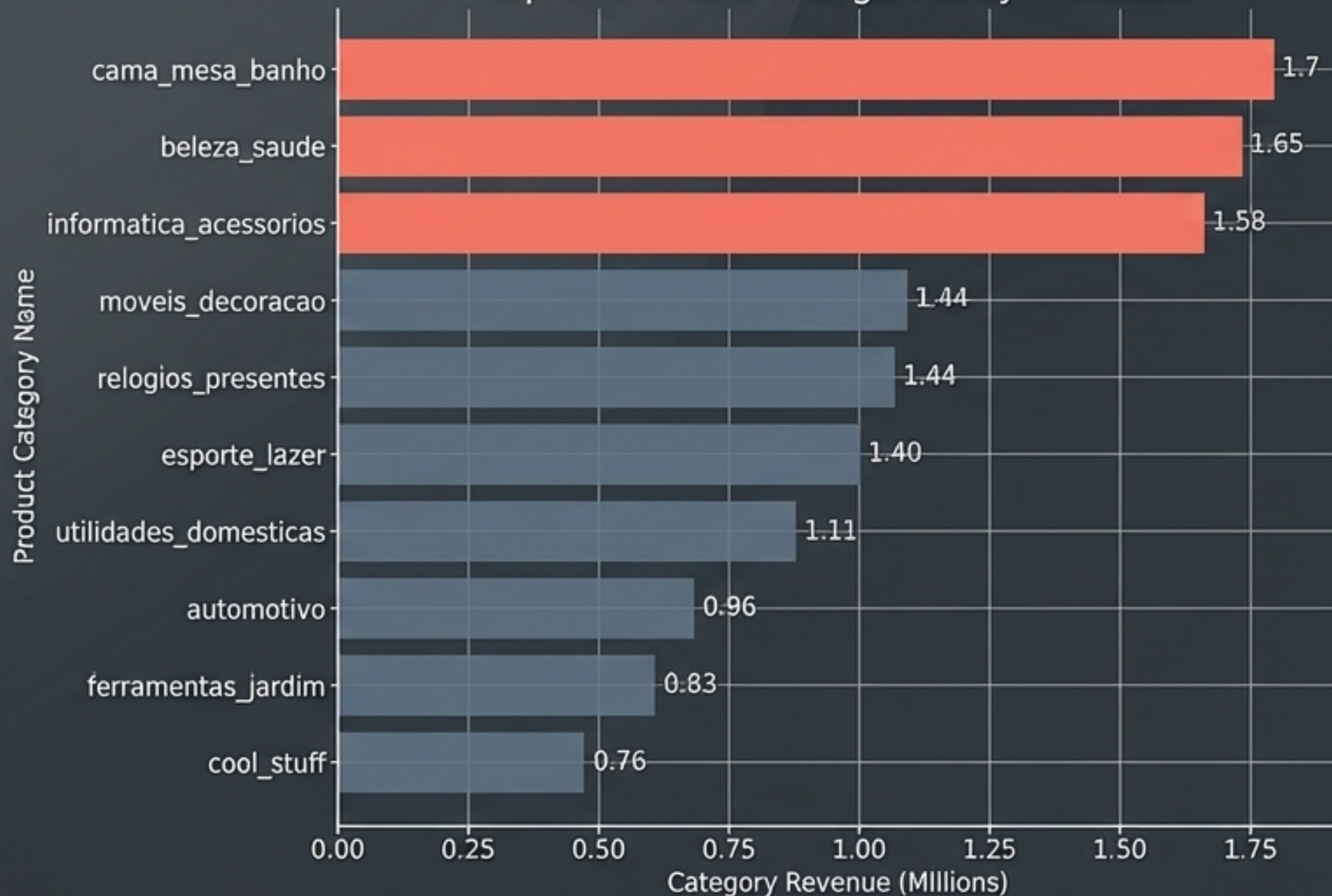
Clear cyclical peaks and troughs are visible throughout the year. The market is not random; it responds predictably to seasonal retail events.

Strategic Implication

The data confirms a highly viable, expanding e-commerce environment capable of sustaining aggressive new market entrants.

Micro Drivers: The Primary Engines of Cash Flow

Top 10 Product Categories by Revenue



The Big Three

'Cama Mesa Banho' (Bed/Bath), 'Beleza Saude' (Health/Beauty), and 'Informatica Acessorios' (Tech Accessories) dominate historical revenue generation.

Consumer Psychology

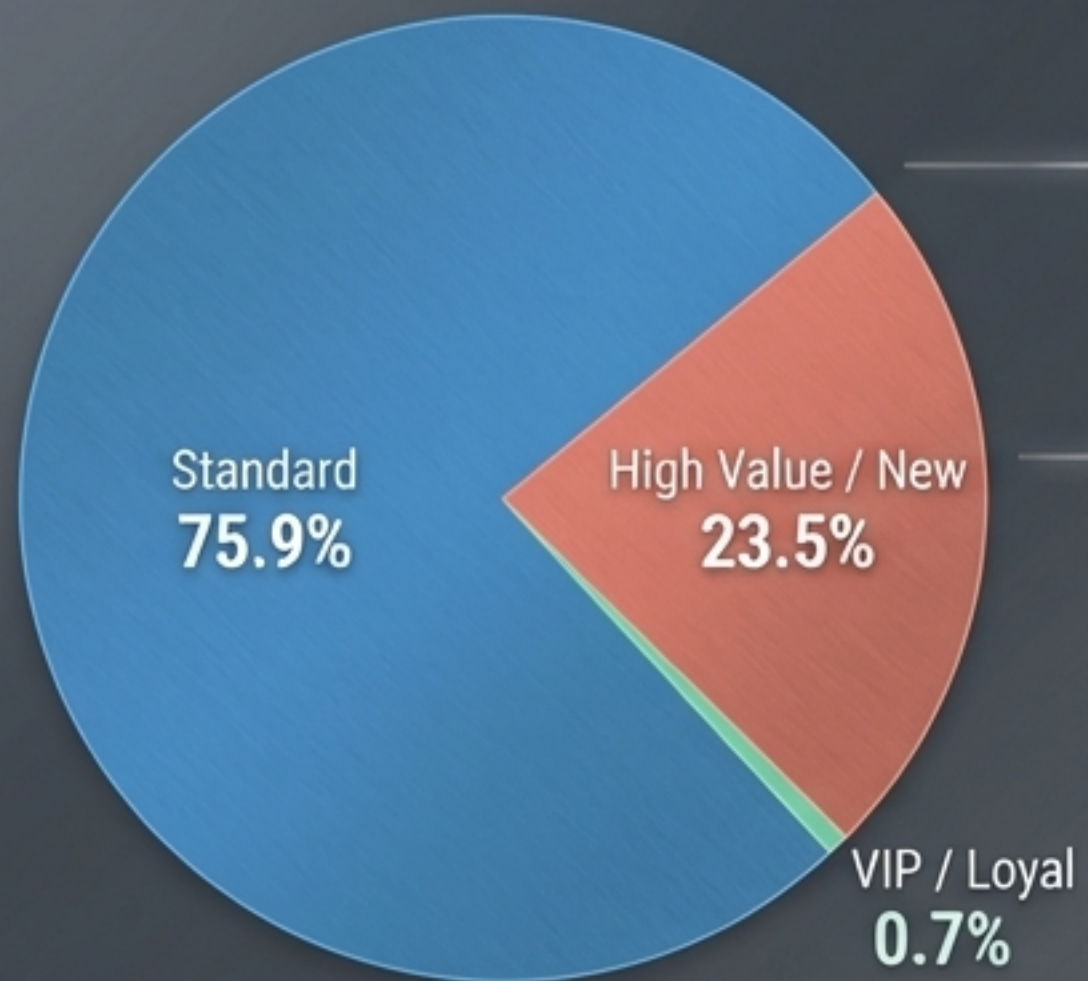
These categories represent a mix of household essentials, personal care, and high-margin technology—indicating sustained, recession-resistant consumer demand.

Strategic Implication

A new venture should not attempt to boil the ocean. Focusing initial capital and inventory on these proven, high-velocity categories drastically reduces market entry risk.

Audience Behavior: The Untapped Loyalty Funnel

Customer Segmentation Distribution



75.9% Standard

The massive top-of-funnel. These are one-off or low-value purchasers.

23.5% High Value / New

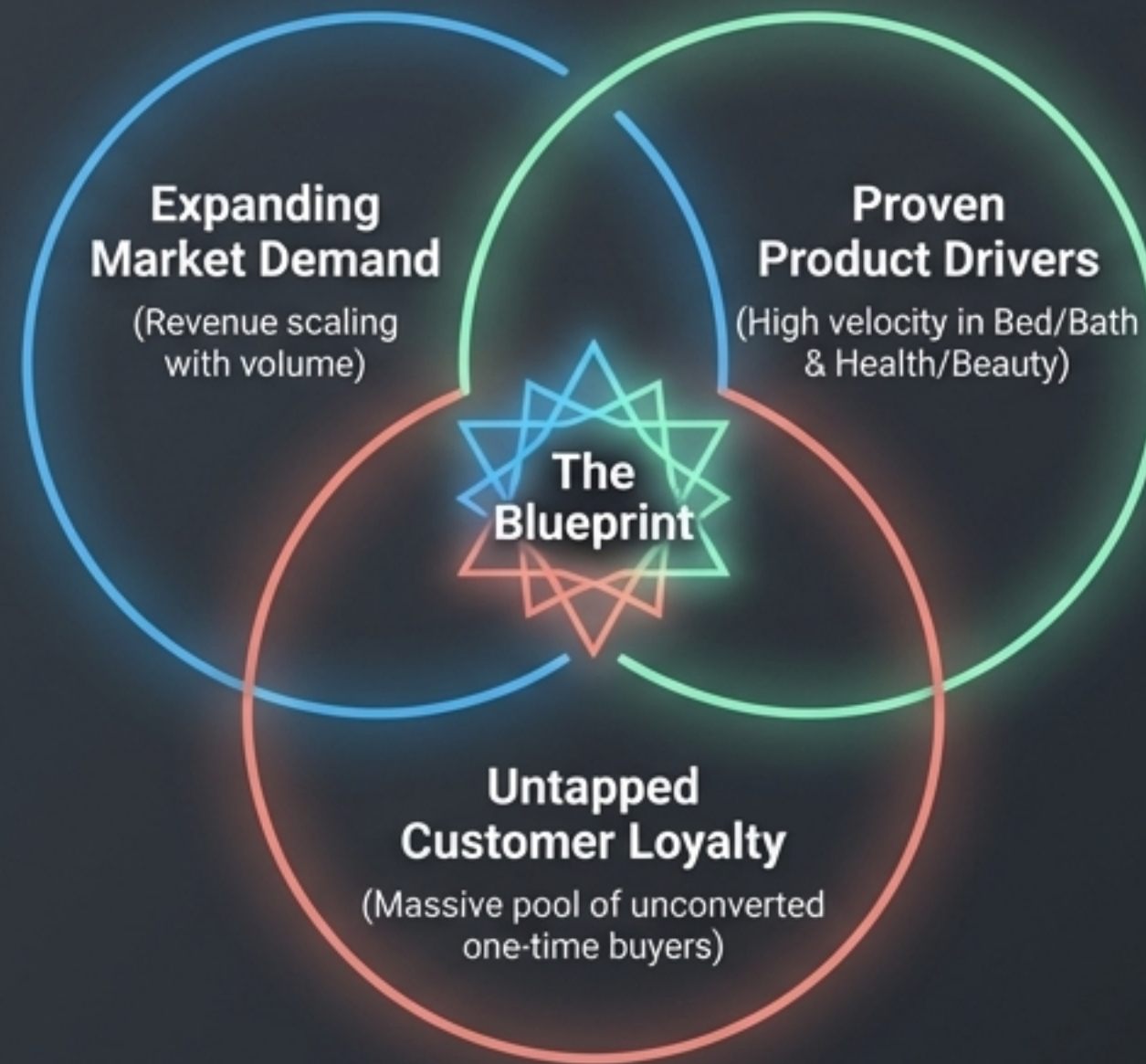
The primary conversion target. They have made a high-value initial purchase but lack frequency.

0.7% VIP / Loyal

The highly lucrative, established core. Small in number, massive in lifetime value.

Core Insight: The e-commerce platform successfully acquires customers at scale, but struggles with retention. The greatest immediate ROI opportunity lies in building mechanisms to transition the 23.5% 'High Value/New' segment into the 'VIP' segment.

Synthesizing the E-Commerce Equation



High Market Demand + Niche Product Focus + Untapped Customer Loyalty = The Blueprint for a New Venture.

Success requires targeting proven categories while competing exclusively on customer retention and post-purchase experience.

Strategic Playbook: Launching a Data-Backed Venture



Pillar 1: Niche Down

Focus: Dominate specific verticals within Health/Beauty or Bed/Bath.

Tactic: Instead of generic home goods, launch with specialized, high-margin niches (e.g., sustainable bedding or organic skincare) to differentiate immediately from massive generalist competitors.



Pillar 2: The Conversion Engine

Focus: Transition the 23.5% 'New' to 'Loyal'.

Tactic: Implement aggressive post-purchase engagement workflows and introduce subscription models for consumable health/beauty products to lock in recurring revenue.



Pillar 3: Operational Agility

Focus: Supply chain built for reality.

Tactic: Leverage the predictable seasonality identified in the sales trends to optimize inventory loading, ensuring peak stock availability right before historical buying waves.

Honest Assessment: Risks and Proactive Mitigations

Technical Track

Risk: Data Drift & Scaling Costs

Raw source formats change, or seasonal data spikes inflate BigQuery computing costs unexpectedly.



Mitigation: Implement incremental dbt loads instead of full refreshes, and deploy automated data contracts to flag schema changes before they break the pipeline.

Business Track

Risk: Supply Chain Bottlenecks

Heavy reliance on top physical categories (Bed/Bath) exposes the venture to logistics failures during seasonal peaks.



Mitigation: Utilize Geographic Dimension data to map highest-density customer zones and pre-position inventory in regional 3PL hubs before peak seasons.

Raw Data is a Commodity. Refined Intelligence is a Competitive Advantage.

The Olist data pipeline provides more than historical reporting—it provides a scalable, rigorously tested engine for future decision-making. We have the architecture to scale, the data quality to trust, and the playbook to win.

End of Briefing. Floor open for Technical and Strategic Discussion.